ConstructionSafety-FaithBench: Auditing Visual-Evidence Faithfulness in Construction-Safety Vision-Language Models

Abstract

Construction-safety vision-language models (VLMs) are being explored as tools for reviewing site imagery, but standard evaluation usually asks only whether the final answer or rationale sounds correct. This leaves a critical gap: a model may predict a safety violation while grounding on irrelevant scene context, or keep the same answer after the rule-relevant worker, protective equipment, or hazard region is removed. We address this gap with ConstructionSafety-FaithBench, a reproducible audit pipeline that separates answer correctness, visual evidence localization, and counterfactual sensitivity. The artifact contains 163 pilot image-rule pairs, 588 scale-up candidates, four safety-rule families, deterministic baselines, a Florence-2 grounding adapter, 948 targeted and matched-random occlusion runs, and a 120-row final audit-label layer. The audit set prioritizes Florence/bootstrap disagreements and weak evidence overlap. Against final audit labels, Florence grounding reaches 18.3% accuracy and 12.5% macro-F1, while a metadata-assisted bootstrap reaches 78.3% accuracy, showing that scaffold-level labels can mask visually grounded failure. Targeted evidence occlusion changes Florence answers 39.2% of the time versus 9.2% for matched random masks, a paired gap of 30.0 percentage points. The final audit labels combine 108 role-conditioned A/B consensus rows with 12 returned adjudication decisions, so results are reported with explicit provenance rather than as unqualified human ground truth.